

AI-Augmented Credit Risk Portfolio Analysis

1. Executive Summary

This report presents a credit risk analysis of a retail loan portfolio, focusing on portfolio health, risk segmentation, and early warning indicators of default. The analysis integrates SQL-based portfolio analytics with an AI-driven probability of default (PD) model to provide both retrospective and forward-looking risk insights.

The portfolio demonstrates an overall default rate of approximately **7.8%**, indicating a broadly stable credit environment. However, risk is unevenly distributed, with a significant concentration of defaults within higher-risk customer segments.

Delinquency (Days Past Due) is identified as a key early warning indicator, while the AI model enhances risk identification by improving precision in isolating high-risk customers. These insights support targeted monitoring and proactive risk management actions.

2. Business Context & Scope

This analysis covers a UK-based retail credit portfolio consisting of unsecured personal loans, auto loans, and mortgage products. All values are reported in GBP (£).

Risk assessment is conducted at a customer-loan level, with a forward-looking horizon of **6–12 months**. Outputs are intended to support risk monitoring and decision-making, not automated credit approval or rejection.

3. Data Overview & Assumptions

The dataset includes customer financial, behavioural, and loan-level attributes such as income, credit score, loan characteristics, delinquency status, and prior default history.

The dataset contains:

- Missing values (e.g. income, employment length)
- Outliers (e.g. high loan/income values)

These reflect real-world data challenges. As the dataset is synthetic, results are indicative and do not incorporate macroeconomic factors.

4. Portfolio Risk Overview

Overall Portfolio Health

The portfolio has:

- **Total Customers:** ~1,000
- **Total Exposure:** ~£13M
- **Default Rate:** ~7.8%
- **Average PD:** ~0.07

This suggests a generally stable portfolio with a strong performing base.

Risk Distribution and Segmentation

Default rates show clear differentiation:

- **High Risk:** ~17%
- **Medium Risk:** ~5%
- **Low Risk:** ~2–3%

This confirms that risk is **concentrated within higher-risk segments**, with High Risk customers contributing disproportionately to defaults.

Delinquency and Early Warning Indicator

- Majority of customers are **Current (DPD = 0)**
- Risk increases significantly for **DPD > 60 days**

Customers in higher delinquency buckets show **materially higher default probabilities**, confirming DPD as a key early warning metric.

Summary

The portfolio appears stable overall; however, risk concentration in higher-risk and delinquent segments highlights the need for targeted monitoring.

5. Risk Segmentation & Early Warning Signals

Risk Segmentation Overview

Customers are segmented into **Low, Medium, and High Risk** based on:

- Delinquency
- Credit score
- Prior defaults

High Risk customers represent the **primary source of credit risk exposure**.

Risk Concentration and Behavioural Patterns

High Risk customers show significantly elevated default behaviour

- Medium Risk customers (~5%) represent a **transition segment**
- Low Risk customers show strong repayment stability

Delinquency as an Early Warning Indicator

Customers in:

- **30–59 DPD → Early risk stage**
- **60+ DPD → High default likelihood**

This highlights a key window for **early intervention**.

Emerging Risk Signals

- Movement into early delinquency (30–59 DPD)
- Concentration of defaults among already stressed borrowers
- Persistence of high-risk behaviour

Implications

- Prioritised monitoring of High-Risk customers
- Early intervention in Medium Risk segment
- Continuous DPD tracking

6. AI Risk Model Overview

A **logistic regression model** was developed to predict Probability of Default (PD).

Model performance:

- **ROC-AUC:** ~0.73 → Good discrimination

Key drivers:

- Previous defaults
- Days past due
- Interest rates
- Loan-to-income ratio

The model improves risk assessment by capturing combined and non-linear effects across variables.

7. Key Findings & Recommendations

Key Findings

- Credit risk is highly concentrated in High-Risk customers
- Delinquency is the strongest predictor of default
- Medium Risk segment (~27% AI default rate) is a critical early-stage group

AI Model Insights

AI-based segmentation shows:

- **High Risk:** ~43% default
- **Medium Risk:** ~27%
- **Low Risk:** ~6%

This demonstrates **improved precision**, with AI identifying a smaller but more concentrated high-risk group.

Recommendations

- Enhance monitoring of High-Risk customers
- Introduce early warning triggers based on DPD
- Target Medium Risk segment for proactive intervention
- Use AI scores to refine risk prioritisation

8. Limitations & Governance Considerations

- Point-in-time dataset (no macroeconomic factors)
- Synthetic data used
- Model not production-calibrated